Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	10 April 2026
Team ID	
Project Name	SHOPEZ : E-commerce Application
Maximum Marks	4 Marks

Technical Architecture:

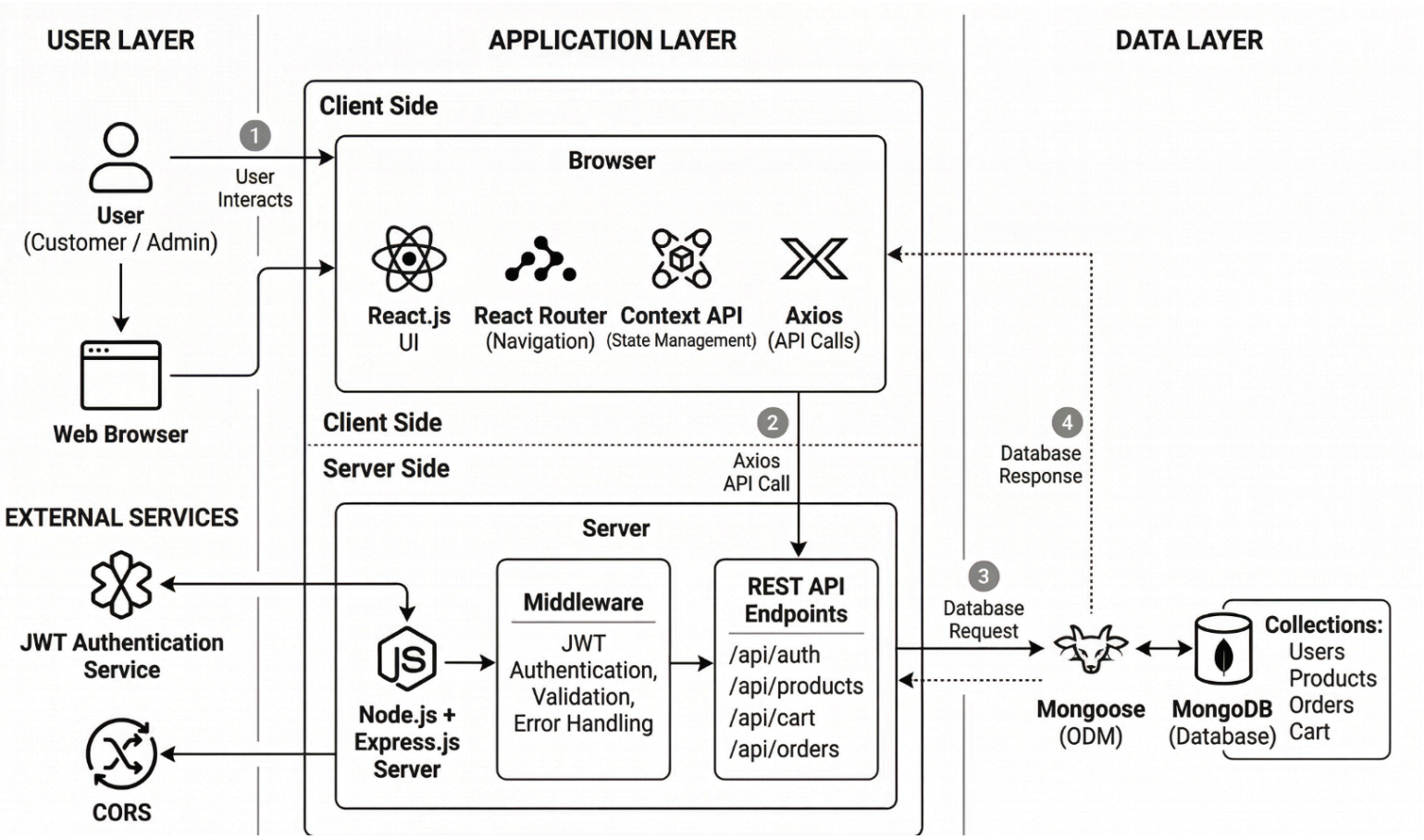

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1.	User Interface	Web-based UI for users to browse products, cart, and orders	React.js, HTML, CSS, Bootstrap
2.	Application Logic-1	Frontend logic for UI rendering and state management	React Context API, React Hooks
3.	Application Logic-2	Backend API handling business logic (auth, cart, orders)	Node.js, Express.js
4.	Application Logic-3	Middleware for authentication, validation, error handling	JWT, express-validator
5.	Database	Stores users, products, orders, cart data	MongoDB (NoSQL), Mongoose
6.	Cloud Database	Cloud-hosted database service	MongoDB Atlas
7.	File Storage	Storage of product images and static files	Local File System / Cloud Storage
8.	External API-1	API communication between frontend & backend	Axios (HTTP Client)
9.	External API-2	Cross-origin communication handling	CORS
10.	Infrastructure (Server / Cloud)	Application deployment environment	Localhost (Dev), Cloud (Node Hosting), GitHub

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Full-stack JS frameworks used for development	React.js, Node.js, Express.js, MongoDB
2.	Security Implementations	Secure authentication and data protection	JWT, bcryptjs, Input Validation
3.	Scalable Architecture	Follows modular and scalable architecture	MVC Architecture, REST API
4.	Availability	Ensures system availability and uptime	Cloud Hosting, MongoDB Atlas
5.	Performance	Optimized request handling and fast UI rendering	Virtual DOM (React), Non-blocking I/O (Node.js), Axios

References:

https://www.mongodb.com/mern-stack

https://expressjs.com/

https://jwt.io/introduction

https://restfulapi.net/

https://c4model.com/

https://nodejs.org/en/docs

https://react.dev/